Semisupervised Spectral Learning With Generative Adversarial Network for Hyperspectral Anomaly Detection

Kai Jiang, Weiying Xie, *Member, IEEE*, Yunsong Li, Jie Lei, *Member, IEEE*, Gang He, and Qian Du, *Fellow, IEEE*

***Abstract*— Limited by the anomalous spectral vectors in unlabeled hyperspectral images (HSIs), anomaly detection methods based on background distribution estimation often suffer from the contamination of anomalies, which decreases the estimation accuracy and, thus, weakens the detection performance. To address this problem, we proposed a novel semisupervised spectral learning (SSL) for the hyperspectral anomaly detection framework based on the generative adversarial network (GAN). GAN is applied and developed to estimate the background distribution in a semisupervised manner and obtain an initial spectral feature because of its strong representational capability and adversarial training advantage. In the proposed framework, an initial spatial feature is generated via morphological attribute filtering. Finally, an exponential constrained nonlinear suppression fusion technique is adopted to suppress the background and combine the complementary information in different features to obtain a fused detection map. The performance of the proposed anomaly detection technique is evaluated on a series of HSIs. Experimental results demonstrate that our method can outperform state-of-the-art anomaly detection methods.**

***Index Terms*— Anomaly detection, background distribution estimation, generative adversarial network (GAN), hyperspectral image (HSI), semisupervised learning.**

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I. Introduction

With the advances of spaceborne and airborne hyperspectral imaging technology, hyperspectral remote sensing has drawn broad attention in recent years. Abundant spectral, spatial, and temporal information can be obtained via a hyperspectral imaging sensor simultaneously [1], [2]. Due to the distinct electromagnetic radiation emitted, reflected, and absorbed by different areas of the Earth's surface based on their unique physical and chemical properties, discriminative spectral information, in a hyperspectral image (HSI), can be employed to identify different materials [1]. With such rich spectral, spatial, and temporal information, HSIs have been widely applied in many fields, such as mineral exploration, environmental monitoring, precision agriculture, and national defense [1], [3].

Based on the recent researches [4], hyperspectral anomaly detection identifies the objects that are significantly different, spatially or spectrally, from their surrounding background without any prior knowledge of background and targets, which has been regarded as one of the most vibrant fields of research in the hyperspectral community and has received considerable interest in the remote sensing field. Anomaly detection cannot be regarded as a binary classification problem in an unsupervised way because the size of the anomaly class is significantly smaller than that of the background class.

To address the aforementioned problem, various anomaly detection methods have been proposed over the past years. Among them, the Reed–Xiaoli (RX) algorithm [5] is regarded as a well-known statistical modeling-based technique, which models the background distribution as a multivariate Gaussian distribution model, whose mean and covariance matrix can be approximated by the undertesting HSI. A series of probabilities that each pixel belongs to the anomaly class can be generated by calculating the Mahalanobis distance between the pixel undertesting and the approximated background mean vector. The RX method has a simple mathematical model and good performance on some data sets. However, in real applications, the distribution of the background is complex and not coincident with the assumption of RX. Thus, several modified RX-based algorithms had been proposed, such as local RX (LRX) [6], subspace RX (SSRX) [7], and kernel-RX (KRX) [8]. The LRX estimates local statistical characters with a local window to

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improve the detection performance and avoid the interference of complex background to a certain extent. In the SSRX, principal component analysis (PCA) [9] is applied to the HSI, and the background is modeled with representative principal components of the HSI. The KRX is a typical nonlinear version of the RX. Alternatively, a collaborative representation-based detector (CRD), which assumes that background pixels can be approximately represented by a linear combination of its spatial neighbors while anomalies cannot, was proposed by Li and Du [10]. Recently, a series of low-rank-based models have been proposed in [11]–[13] under the assumption that the background space is low rank and the anomaly space is sparse. A few methods, such as robust PCA (RPCA) [14], go decomposition (GoDec) [15], and low-rank representation (LRR) [16], have been employed to decompose HSI for detecting anomalies from background and noise. Ma *et al.* [17] proposed a novel hyperspectral anomaly detection method by designing a multiple-dictionary sparse-feature-extraction approach. In addition, Kang *et al.* [18] proposed a novel Attribute and Edge-preserving filtering-based Detection algorithm (AED) that can be implemented efficiently and achieve a competitive detection capability although its false alarm rate needs to be further reduced. More recently, a series of deep learning-based hyperspectral anomaly detection algorithms have been proposed. Li *et al.* [19] proposed a novel HSI anomaly detection framework with transferred deep convolutional neural network (CNN). A novel HSI anomaly detection algorithm based on CNN and LRR is proposed in [20]. Xie *et al.* [21] proposed a spectral adversarial feature learning (SAFL) architecture for hyperspectral anomaly detection to better distinguish between the background and anomalies. Jiang *et al.* [22] proposed a generative adversarial network (GAN)-based detection framework [23] to learn a discriminative background reconstruction with anomaly targets being suppressed, such that an initial detection result can be generated by the residual between the original and reconstructed images.

Deep neural network (DNN)-based anomaly detection methods [24], which can be divided into three categories (i.e., supervised, unsupervised, and semisupervised [25] anomaly detections) according to the extent of availability of labels, have also been proposed in various fields, such as illegal traffic flow detection [26], retinal damage detection [27], and cyber-network intrusion detection [28], [29]. Supervised methods can be regarded as training a supervised binary or multiclass classifier with labels of both normal and anomalous training instances [24]. Despite the superior performance of supervised methods [30], these methods are not as applicable as unsupervised or semisupervised methods due to the lack of availability of labeled training samples. In contrast, since the labeled data are very hard to obtain in some situations, unsupervised methods are adopted to learn useful representations for unlabeled data and detect anomalies based on intrinsic properties of the data instances. For example, [31] proposed an unsupervised HSI anomaly detection method, in which the reconstruction errors between the input and the output of the deep belief network (DBN) [32] and autoencoder (AE) [33] are adopted as the anomaly score directly. However, unsupervised techniques are very sensitive to noise and data corruption and are often less accurate than supervised techniques due to lack of the constraint from the labeled training instances [24]. Based on the observation that the labels of anomalies are far more difficult to obtain than normal instances, in order to approximate the performance of supervised methods while relaxing the limitation of lack of availability of labeled training samples, semisupervised methods are adopted in anomalies field [34]–[36], which only leverages existing labels of normal training samples to train a model, and the model is then applied to both normal and anomalous samples to distinguish anomalies from all the samples. Akcay *et al.* [37] proposed a novel conditional GAN-based anomaly detection model (as shown in Fig. 1) that jointly learns the generation in high-dimensional image space and the features in a latent space. In this sense, semisupervised methods can be considered as a tradeoff between the requirements of labeled instances and detection performance. However, in hyperspectral anomaly detection, both the labels of normal and abnormal instances are unavailable, which limits the application of supervised methods. Meanwhile, unsupervised methods are very sensitive to anomalous spectral vectors and have the risk of overfitting to anomalies that would weaken the detection performance. In summary, the existing supervised and unsupervised anomaly detection methods cannot deal with the problem of insufficient samples very well.

In this article, we propose a novel GAN-based semisupervised spectral learning (SSL) for hyperspectral anomaly detection to fully explore the complementary background information among spectral and spatial features of HSI. The proposed hyperspectral anomaly detection framework comprises three critical steps: background estimation with SSL, spatial-feature-extraction, and feature fusion. In order to implement the SSL for background distribution estimation on the unlabeled original HSI without any prior, an adaptive algorithm termed coarse screening and sample selection is proposed to reject the instances with high likelihoods of being anomalous and accept the instances with high likelihoods of being normal to automatically construct a relatively pure background spectra data set. It is assumed that anomalous spectra are with low probability, low statistical energy, and scattered in the spatial domain. In the background estimation procedure, GAN is adopted and developed to estimate the background distribution and obtain the spectral domain feature using its strong representational capability and adversarial training advantage. In order to further stabilize the representation learning and enhance the accuracy of background distribution estimation, two powerful constraints, i.e., a continuity representation constraint in the latent space and a discriminator-based authenticity constraint, are also utilized to further ensure that background distribution is reliably and stably estimated, and the anomalies are poorly represented. As a result, the residual between the original HSI and reconstructed background is the anomaly likelihood for the pixels of HSI in the spectral domain. For the spatial-feature-extraction procedure, first, PCA is applied to extract the principal component of HSI. Then, the morphological attribute filter is adopted to detect objects with specific area properties and to generate an initial spatial feature. Finally, an exponential

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constrained feature fusion procedure is applied to fuse the above-mentioned two features and obtain the final detection result, aiming at avoiding the limitations of only detecting from a single perspective (i.e., the spectral domain or the spatial domain) and further increasing detection performance. The major contributions of this article are summarized as follows.

1) An adaptive algorithm, which automatically accepts the instances with a high likelihood of being normal and constructs a relatively pure background data set without any prior, is proposed to address the issue of lacking training instances for semisupervised learning.
2) A semisupervised GAN-based model, which relieves the distribution assumptions requirement during the background distribution estimation, is proposed to effectively improve the background modeling accuracy and contribute to detection performance.
3) An exponential constrained feature fusion procedure is proposed, in which the combination of the complementary information of spectral and spatial features makes a contribution to background suppression and leads to satisfactory detection performance. We show that the proposed method is superior to the existing state-of-the-art methods.

The remainder of this article is organized as follows. A review of related works is given in Section II. Section III describes the details of the proposed anomaly detection framework. The experimental results and analysis are presented in Section IV, which is followed by the conclusion and future works in Section V.

II. Related Works

GAN [23], [38], which provides a way to learn deep representations without extensively annotated training data in a semisupervised or unsupervised manner, was first introduced by Goodfellow *et al.* [23]. As discussed in [39], GAN consists of two subnetworks: generator G and discriminator D, both of them are implicit mapping expressions that are usually implemented by DNNs [40]. The structure of GAN is shown in Fig. 1, where G intends to generate as real a sample as possible from a prior distribution, while D receives both fake (generated by G) and real (authentic) samples and aims to tell them apart. The two subnetworks compete against each other and eventually reach a Nash equilibrium. Thus, the high-dimensional distribution of data can be implicitly modeled by training a pair of networks (i.e., generator G and discriminator D) in competition with each other.

Let $\mathbf{z}$ be a random variable from the prior distribution $p(\mathbf{z})$. The generated sample is recorded as $G(\mathbf{z})$, whose distribution is recorded as $p_g(\mathbf{z})$. The objective of GAN is to make $p_g(\mathbf{z})$ as similar as possible to the distribution of training data set $p_r(\mathbf{x})$. A probability or a scalar $D(G(\mathbf{z}))$, which estimates whether a sample is from real distribution, is generated by the discriminator D. During the training procedure of GAN, the parameters of G and D are updated iteratively. The loss

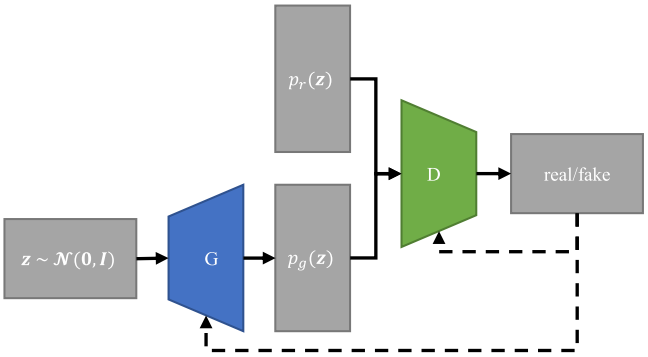


Fig. 1. Basic framework of GAN, which includes a generator G and a discriminator D, is trained through the backpropagation algorithm.

function of GAN can be formulated as

$$\min_G \max_D E_{\mathbf{x}\sim p_r(\mathbf{x})}\left[\log D(\mathbf{x})\right] + E_{\mathbf{z}\sim p_g(\mathbf{z})}\left[\log(1 - D(G(\mathbf{z})))\right]. \tag{1}$$

In real applications, the alternate minimization strategy is generally adopted to update the parameters of generator G and discriminator D. However, as discussed in [41], GAN had been plagued by convergence difficulties, failing of indicating the training process and mode collapse. Since then, a series of researchers [41], [42] had tried to solve the aforementioned problems but with few satisfactory results. In [41], a series of techniques, including feature matching and minibatch discrimination, were proposed to stabilize the convergence of GAN. In [42], model collapse is analyzed, and a modified discriminator is proposed to alleviate the problem of model collapse. However, none of the above-mentioned methods completely solve the problems. Finally, Tolstikhin *et al.* [43] analyzed the problem of the original GAN theoretically, thus, pointed out the key for improvement, and proposed the Wasserstein GAN (WGAN), which successfully achieved the following points.

1) The instability of the training procedure has been thoroughly avoided.
2) A solution is proposed to improve the quality of generated results and to solve the collapse mode problem.
3) A metric is utilized to indicate the progress of GANs convergence.

Thus, from the state-of-the-art GAN architecture proposed by Tolstikhin *et al.* [43], WGAN is adopted to learn a representation of background variability in a semisupervised manner in this article.

III. Proposed Method

Fig. 2 illustrates the schematic of the proposed GAN-based SSL hyperspectral anomaly detection framework, which consists of two critical components: SSL for background distribution estimation and spatial feature extraction. For the former component, a GAN-based SSL model with two constraints is proposed to estimate the background distribution, and a spectral feature can be inferred by applying the trained model to all the spectral vectors. Thus, the spectral feature, in which each pixel represents a likelihood of being anomalous corresponding to a spectral vector of the HSI, can be yielded.

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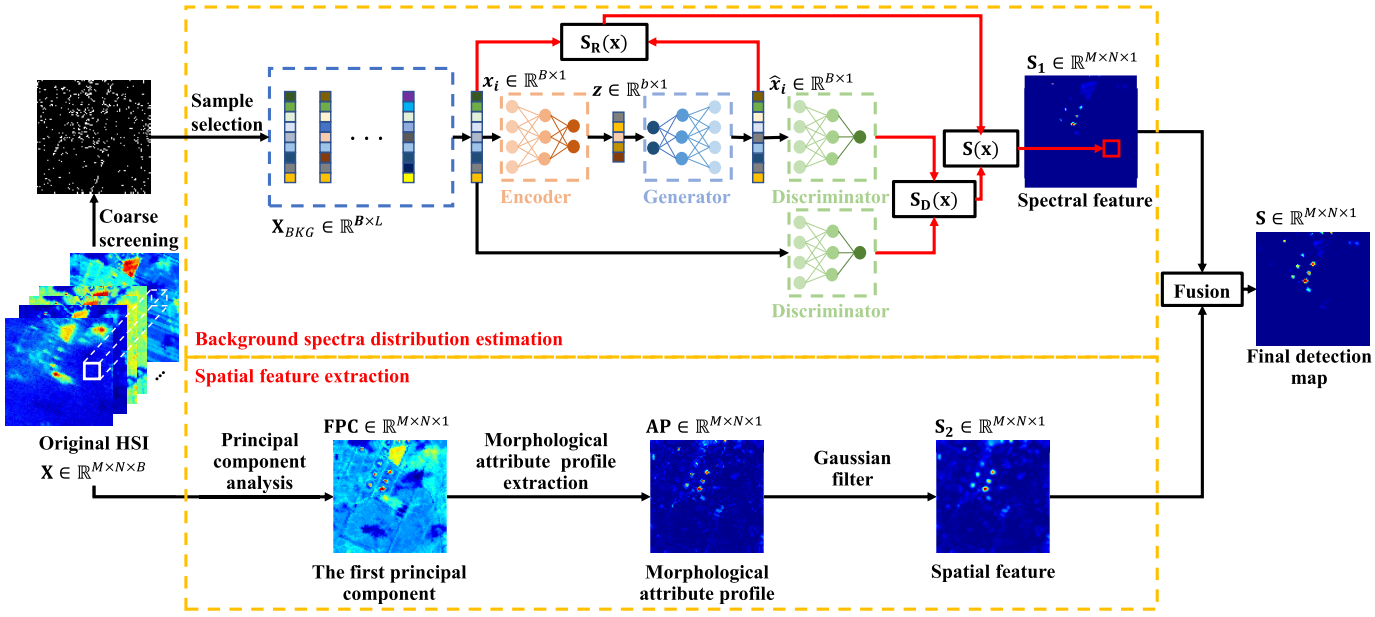


Fig. 2. Schematic of the proposed hyperspectral anomaly detection framework.

For the latter, potential anomalies, which appear with specific area properties, can be detected with a morphological attribute filter. Finally, an exponential constrained nonlinear feature fusion component is applied to fuse the above-mentioned two features to produce the final detection map.

A. SSL for Background Distribution Estimation

In this article, a GAN-based SSL model, which learns features from data sets with unknown distribution and estimates the data distribution without any prior knowledge, is adopted to estimate the background distribution in a semisupervised manner. In this way, the requirement of the background distribution assumption can be avoided. Moreover, two powerful constraints, i.e., a continuity representation constraint in the latent space and a discriminator-based authenticity constraint, are utilized to ensure that the background distribution is reliably estimated and the anomalies are poorly represented. As a result, the residual between the original HSI and reconstructed background is regarded as the anomaly likelihoods for the pixels of HSI in the spectral domain.

However, the training data set used to estimate background distribution may be contaminated by anomalies. In order to decrease the anomalous pixel contamination, an adaptive algorithm termed coarse screening and sample selection is applied to preliminarily reject the instances with the high likelihood of being anomalous and accept the instances with the high likelihood of being normal to automatically construct a relatively pure background spectra data set. However, because of the limitations of the coarse screening and sample selection procedure, the constructed background spectral data set would still inevitably contain a few anomalous samples. Based on the fact that anomalies are far less than the normal instances in the constructed background spectral data set, two widely used training strategies, i.e., limiting model capacity [44] and dropout [45], are applied to avoid the model overfitting anomalies and provide a more reliable background estimation.

The training procedure of the proposed hyperspectral anomaly detection framework is shown in Fig. 3. More details of the training procedure and strategies will be described in the following sections.

1) Coarse Screening and Sample Selection: As shown in Fig. 2, an adaptive algorithm, which is termed coarse screening and sample selection, is proposed to construct a relatively pure background spectra data set. Based on the observation that the objects with small area properties and distinguished energy in the spatial domain always have a high likelihood of being anomalies, morphological opening and closing [46] with a small size of the structuring element are applied on the first principal components of HSI to extract an initial likelihood map, in which each pixel represents the likelihood of a certain spectrum being anomalous. Later, a binary map, i.e., the coarse screening map, can be generated by applying the Otsu's thresholding algorithm [47] to the extracted initial likelihood map. A relatively pure spectral data set can be constructed by rejecting the samples with a high likelihood of being anomalous with the guidance of the coarse screening map. Thus, with the help of the coarse screening, the potential anomalous samples are preliminarily rejected, and a relatively pure background spectral data set $\mathbf{X}_{\mathrm{BKG}} \in \mathbb{R}^{B\times L}$ with L spectra and B spectral bands can be constructed, which is used to the SSL for the background distribution estimation.

2) Semisupervised Learning for Background Distribution Estimation: Let $\mathbf{X} \in \mathbb{R}^{M\times N\times B}$ denote the original HSI data with $M \times N$ pixels and B spectral bands acquired by an optical sensor and registered over some geographical area. $\mathbf{X}_{\mathrm{BKG}} \in \mathbb{R}^{B\times L}$ refers to the constructed background spectral data set in Section III-A1. The background spectra vectors $\mathbf{x}_i \in \mathbb{R}^{B\times 1}(i = 1, 2, I, L)$ sampled from the data set $\mathbf{X}_{\mathrm{BKG}}$ are adopted as the training data for the GAN-based

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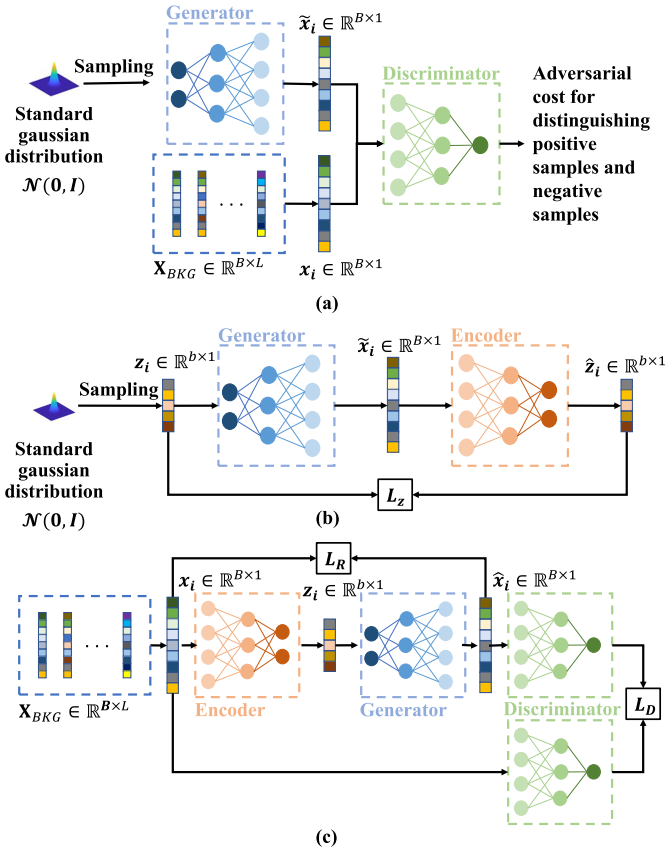


Fig. 3. Training procedure of the proposed hyperspectral anomaly detection framework. (a) GAN-based learning. (b) lvl training procedure. (c) vlv training procedure.

SSL model to learn the distribution of background spectral vectors in a semisupervised manner. During the evaluation stage, all the spectral vectors in the original HSI $\mathbf{X}$ are fed into the trained model to generate a residual between the original HSI and reconstructed background, which represents the anomaly likelihood for each pixel of HSI in the spectral domain.

As shown in Fig. 2, the generator G in the GAN model tries to learn a distribution p_d, which is an approximation of the real background distribution p_x. A random noise vector z sampled from a prior distribution, which is generally selected as the standard Gaussian distribution or uniform distribution, is fed into G, and the output of G, $G(\mathbf{z})$, is expected to obey distribution p_x and to exhibit similar spectral properties to the normal spectral vectors. In this setting, a fully connected neural network parameterized by θ is adopted as the architecture of generator G to perform the nonlinear mapping $G(\mathbf{z}; \theta)$. The input of discriminator is either a spectral vector sampled from $\mathbf{X}_{\mathrm{BKG}}$ or a sample generated by the generator G. The output of discriminator, which is a single scalar value, can be interpreted as the probability of the input being a real spectral vector $\mathbf{x}_i$ or a generated sample $G(\mathbf{z}; \theta)$. Thus, the discriminator is a fully connected neural network parameterized by ϕ that maps a spectral vector to a single scalar value, i.e., $D(\mathbf{x}; \phi)$. During the GAN training, D and G are optimized simultaneously in an adversarial manner, that is

$$\min_G \max_D E_{\mathbf{x}\sim p_x(\mathbf{x})}\big[\log D(\mathbf{x})\big] + E_{\mathbf{z}\sim N(\mathbf{0},\mathbf{1})}\big[\log(1 - D(G(\mathbf{z})))\big]. \tag{2}$$

The generator G is trained to make its output $G(\mathbf{z})$ obey the distribution p_x and fool the discriminator D, while D is trained to distinguish whether the input is a spectral sampled from the data set $\mathbf{X}_{\mathrm{BKG}}$ or a sample generated by G.

Furthermore, as mentioned earlier, due to the instability of vanilla GAN, problems such as mode collapse and vanishing gradient always exist during the training procedure of GAN, which decreases the estimation accuracy of the background distribution and negatively impact on the downstream task. In order to avoid the aforementioned shortcomings of vanilla GAN and stabilize the training procedure, a modified version, i.e., WGAN, is adopted to estimate the distribution of background distribution. Thus, the optimization objective is modified as [43]

$$\min_{\mathrm{G}} \max_{\mathrm{D}} \big[E_{\mathbf{x}\sim p_x(\mathbf{x})}(D(\mathbf{x})) - E_{\mathbf{z}\sim N(\mathbf{0},\mathbf{1})}(D(G(\mathbf{z})))\big]. \tag{3}$$

As discussed earlier, the constructed background spectral data set $\mathbf{X}_{\mathrm{BKG}}$ would still inevitably contain a few anomalous samples. In order to avoid the model overfitting the anomalies in $\mathbf{X}_{\mathrm{BKG}}$ and provide a more reliable background estimation, both the dropout with probability of 0.5 and the l1-norm regularization constraint with a weighting factor of 2.5×10^{-5} are applied to the generator and the discriminator. Thus, the optimization objective function can be rewritten as

$$\min_{\theta} \big[E_{\mathbf{x}\sim p_x(\mathbf{x})}(D(\mathbf{x})) + \epsilon \cdot g(\theta, \phi)\big] \tag{4}$$

$$\min_{\phi} \big[-E_{\mathbf{x}\sim p_x(\mathbf{x})}(D(\mathbf{x})) + E_{\mathbf{z}\sim N(\mathbf{0},\mathbf{1})}(D(G(\mathbf{z}))) + \epsilon \cdot g(\theta, \phi)\big] \tag{5}$$

where $g(\theta, \phi) = 0.5\|\theta\|_1 + 0.5\|\phi\|_1$ and $\epsilon = 2.5 \times 10^{-5}$.

Finally, the distribution of background can be estimated by the generator, and the fitting degree between the generated samples and real background distribution p_x can be quantified by the discriminator. Both the generator and the discriminator with fixed parameters are applied to downstream tasks.

However, due to the WGAN's strong representation and generalization capability, variants that do not obey the original background distribution may be generated [48]. In order to further stabilize the representation learning and enhance the separation between normal and abnormal categories, as illustrated in Fig. 3, two powerful constraints, i.e., a continuity representation constraint in the latent space and a discriminator-based authenticity constraint, are also imposed to further ensure that background distribution is reliably and stably estimated, and the anomalies are poorly represented.

3) Mapping Learning From Spectral Vector to Latent Space With Two Constraints: After finishing the training procedure of WGAN, a mapping [i.e., $G(\mathbf{z})$] between the standard Gaussian distribution and the distribution of background of HSI is established. Based on the assumption that the latent space has the property of smooth transitions, similar spectral vectors would be generated by two close points sampling from the latent space. Given a testing spectral vector $\mathbf{x}$, a latent

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vector $\mathbf{z}$, which obeys the standard Gaussian distribution, should be yielded based on the criterion that the corresponding generated vector G(z) and spectral vector $\mathbf{x}$ should be as similar as possible. The degree of similarity between $\mathbf{x}$ and $G(\mathbf{z})$ would be adopted as an index to characterize the anomaly likelihood. However, the inverting mapping between the background distribution and latent distribution (i.e., the aforementioned standard Gaussian distribution) has not been established automatically. In addition, due to the lack of constraints in the latent space, variants that do not obey the original background distribution may be generated because of the WGAN's strong representation and generalization capability [48]. In order to further stabilize the representation learning, find the most accurate $\mathbf{z}$ in the latent space, and enhance the accuracy of background distribution estimation, as shown in Fig. 3, we make a proposal to train an encoder $E(\cdot)$ to learn the mapping between $\mathbf{x}$ and $\mathbf{z}$ with two powerful constraints, i.e., a continuity representation constraint in the latent space (which is termed lvl (latent-visible-latent) training procedure in the following) and a discriminator-based authenticity constraint (which is termed vlv (visible-latent-visible) training procedure in the following). As shown in Fig. 3, an AE architecture, in which the encoder $E(\cdot)$ with trainable parameters maps the spectral vectors to the latent space, and the generator $G(\cdot)$ with fixed parameters resulting from WGAN training acts as a decoder to establish the mapping from latent distribution to background distribution, is adopted during the lvl and vlv training procedures. More details about the above-mentioned two encoder training procedures are described as follows.

a) Continuity representation constraint in latent space—lvl training procedure: Based on the fact that the latent space has the property of smooth transitions and similar spectral vectors would be generated by two close points in the latent space, an anomalous instance, which is mapped to a position in the latent space near a point corresponding to a normal instance, would be reconstructed as a normal instance, which means that the residual between the anomalous instance and its reconstructed version is large. Thus, the instance can be detected as anomalous with a high degree of confidence. However, due to the strong reconstruction capability of AE and the lack of continuity constraints in the latent space, it cannot ensure a powerful separation for normal and abnormal instances.

As shown in Fig. 3(b), in order to improve the separation between the normal and abnormal categories and ensure the continuity in the latent space, an AE with a reversed order of encoder and decoder is adopted to formulate a latent continuity representation constraint for the model, i.e., lvl training procedure.

A random vector $\mathbf{z}$ sampled from the latent space is mapped as a spectral vector by the generator $G(\cdot)$ with fixed parameters. Then, the spectral vector can be mapped back to the latent space by the trainable encoder $E(\cdot)$. The mean squared error (MSE) between the input sample $\mathbf{z}$ and the reconstructed sample $E(G(\mathbf{z}))$ is minimized during the lvl training procedure

$$L_{\mathrm{lvl}}(\mathbf{z}) = ||\mathbf{z} - E(G(\mathbf{z}))||_2^2. \tag{6}$$

With this procedure, the output of the encoder $E(\cdot)$ is ensured to obey the standard Gaussian distribution, and the representation consistency constraint can be imposed on the latent space.

b) Discriminator-based authenticity constraint—vlv training procedure: Due to the WGAN's strong representation and generalization capability, variants that do not obey the original background distribution may be generated. Intuitively, MSE between a query spectrum and its reconstructed version can be adopted as a metric to enforce similarity in the spectra space. However, minimizing MSE during the training procedure only enforces the similarity between a spectral vector $\mathbf{x} \in \mathbf{X}_{\mathrm{BKG}}$ and its reconstructed version $E(G(\mathbf{x}))$, but the true target location in the latent space of a given query spectral vector is unknown. The mapping of new instances may result in positions in regions of the latent space only sparsely sampled during background distribution estimation that would not convince the discriminator when mapped back to the spectra space. As a consequence, only minimizing MSE will occasionally yield unrealistic reconstructions of normal samples but still have a small residual even for anomalous instances, which decreases the anomaly detection performance, while we found that the discriminator can be adopted as a reliable basis for identifying unrealistic instances and contribute to detection performance. Thus, the discriminator $D(\cdot)$ of the trained WGAN was introduced to train the encoder with the guidance of discriminator to constrain the authenticity of samples.

As shown in Fig. 3(c), a modified AE architecture with the constraint of the discriminator is proposed to formulate vlv training procedure and further train the encoder $E(\cdot)$. The mapping from a real spectral vector to the latent space is performed by the trainable encoder $E(\cdot)$, and the mapping from latent space is performed by the trained generator $G(\cdot)$ with fixed parameters. During the vlv training procedure, the MSE between spectral vector $\mathbf{x}$ and reconstructed vector $E(G(\mathbf{x}))$, as well as the discriminator-based authenticity constraint term, is minimized

$$L_{\mathrm{vlv}}(\mathbf{x}) = ||\mathbf{x} - G(E(\mathbf{x}))||_2^2 + \lambda \cdot ||D(\mathbf{x}) - D(G(E(\mathbf{x})))||_2^2 \tag{7}$$

where λ is a weighted parameter.

Thus, the discriminator-based authenticity constraint term between the spectral vector $\mathbf{x}$ and its reconstruction version $E(G(\mathbf{x}))$ is introduced to ensure that the model can yield realistic reconstructions with a low MSE for normal samples and unrealistic reconstructions with a high MSE for anomaly samples.

4) Generating the Spectral Feature: During the spectral-feature-extraction procedure, the similarity between the input spectral vector and its reconstruction version is adopted to quantify the spectral feature. The loss function $L_{\mathrm{vlv}}(\mathbf{x})$, which is used for encoder training, is directly selected as the spectral feature $S_1(\mathbf{x})$ for an individual spectral vector $\mathbf{x} \in \mathbf{X}$

$$S_1(\mathbf{x}) = ||\mathbf{x} - G(E(\mathbf{x}))||_2^2 + \lambda \cdot ||D(\mathbf{x}) - D(G(E(\mathbf{x})))||_2^2. \tag{8}$$

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Thus, as shown in Fig. 2, a spectral feature $\mathbf{S}_1$, in which each pixel represents the score of a certain spectrum being anomalous, can be generated by calculating $S_1(\mathbf{x})$ for each spectral vector of an HSI

$$\mathbf{S}_1(\mathbf{X}) = ||\mathbf{X} - G(E(\mathbf{X}))||_2^2 + \lambda \cdot ||D(\mathbf{X}) - D(G(E(\mathbf{X})))||_2^2. \tag{9}$$

B. Spatial Feature Extraction

Potential anomalies always appear with small and specific area properties and distinguished gray value. In order to better detect the anomalies and reduce the false alarm rate, morphological attribute thinnings and thickenings [49] are applied to the first principal components of HSI to produce an attribute profile, i.e., the connected components in a small area. The bright-connected components and the dark-connected components are removed by the morphological attribute thinnings and thickenings, respectively. Thus, the connected components with a small area and distinguished gray value can be extracted as

$$\mathbf{AP} = |\mathbf{FPC} - \delta(\mathbf{FPC})| + |\mathbf{FPC} - \phi(\mathbf{FPC})| \tag{10}$$

where **FPC** is the first principal component of HSI **X**, and $\delta(\mathbf{FPC})$ and $\phi(\mathbf{FPC})$ refer to the result of morphological attribute thinnings and thickenings operations, respectively. Furthermore, (10) can be reformulated as

$$\mathbf{AP} = \phi(\mathbf{FPC}) - \delta(\mathbf{FPC}). \tag{11}$$

In (10), the former term generates the bright-connected components, while the dark-connected components are produced by the latter term. Thus, the morphological attribute profile, i.e., the potential anomalies with small and specific area properties and distinguished gray values, can be generated. The morphological structure information and the spatial feature can be characterized by the extracted morphological attribute profile. In addition, the Gaussian filter is applied to the extracted morphological profile to further reduce the miss rate

$$\mathbf{S}_2 = \mathrm{GF}(\mathbf{AP}). \tag{12}$$

C. Feature Fusion With Exponential Constrained Nonlinear Suppression Function

Complementary information is provided by the two initial features $\mathbf{S}_1$ and $\mathbf{S}_2$ obtained from different aspects (i.e., the spectral domain and the spatial domain, respectively). In addition, as shown in Fig. 2, the initial feature $\mathbf{S}_1$ in the spectral domain always appears with low false alarm and properly background suppression. Thus, an exponential constrained nonlinear transform function based on $\mathbf{S}_1$ is proposed to suppress the background and enhance the detection performance

$$\kappa = \left(1 - e^{-\gamma \cdot \mathbf{S}_1}\right) \tag{13}$$

where γ is a positive parameter to adjust the shape of (13). The instances with high likelihoods of being anomalous in the spectral domain are preserved, and κ is relatively stable and approximates to 1. On the contrary, those with a high likelihood of being normal are suppressed, and κ falls sharply and approximates to 0. Fig. 4 shows the shape of (13) under different choices of γ. Thus, a feature fusion procedure is introduced to suppress the background properly and avoid the limitations of only detecting from a single perspective

$$\mathbf{S} = \kappa \cdot \mathbf{S}_2. \tag{14}$$

As shown in Fig. 2, the background is well suppressed, and the anomaly targets are detected satisfactorily.

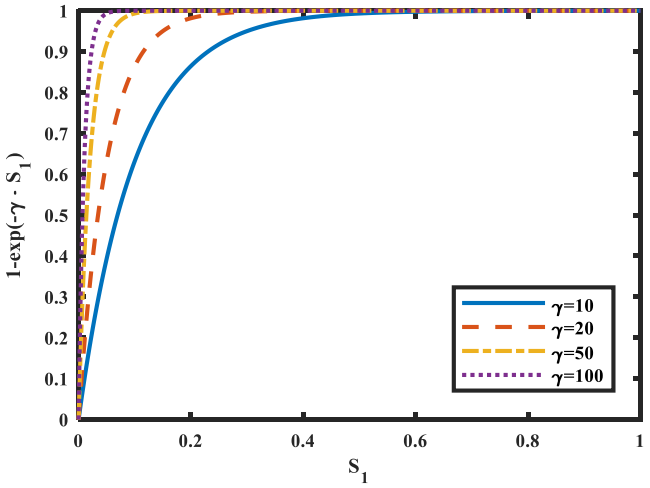


Fig. 4. Shape of exponential constrained nonlinear suppression function $\kappa(\mathbf{S}_1) = 1 - \exp(-\gamma \cdot \mathbf{S}_1)$ with different choices of γ.

IV. Experimental Results and Discussion

A. Data Sets

In this article, the proposed method is evaluated on four real HSIs captured from different real scenes that are listed in Table I.

1) Texas Coast Data Set: It is a series of scenes of the Texas coast area captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. It has two images with 100 × 100 pixels and 207 bands from 0.45 to 1.35 μm. The data set is corrupted by a series of strips noise and is regarded as a challenging data set. Buildings with different scales are adopted as anomalies. The pseudocolor image and reference map are shown in Fig. 5(a) and (b).

2) Los Angeles Data Set: It was acquired by the AVIRIS sensor covering the Los Angeles city area. It has three images with 100 × 100 pixels and 205 spectral bands from 0.43 to 0.86 μm. The objects with different scales are considered as anomalies to be detected. The pseudocolor images and reference maps are presented in Fig. 5(c)–(e).

3) El Segundo Data Set: This data set was captured by the AVIRIS sensor covering the urban scene of El Segundo, which has 250 × 300 pixels and 224 spectral bands varying from 0.37 to 2.50 μm. An area of an oil refinery, several residential areas, parks, and a school zone are contained in the scene, and the construction of the oil refinery, such as storage tanks and the towers, is regarded as the anomalies to

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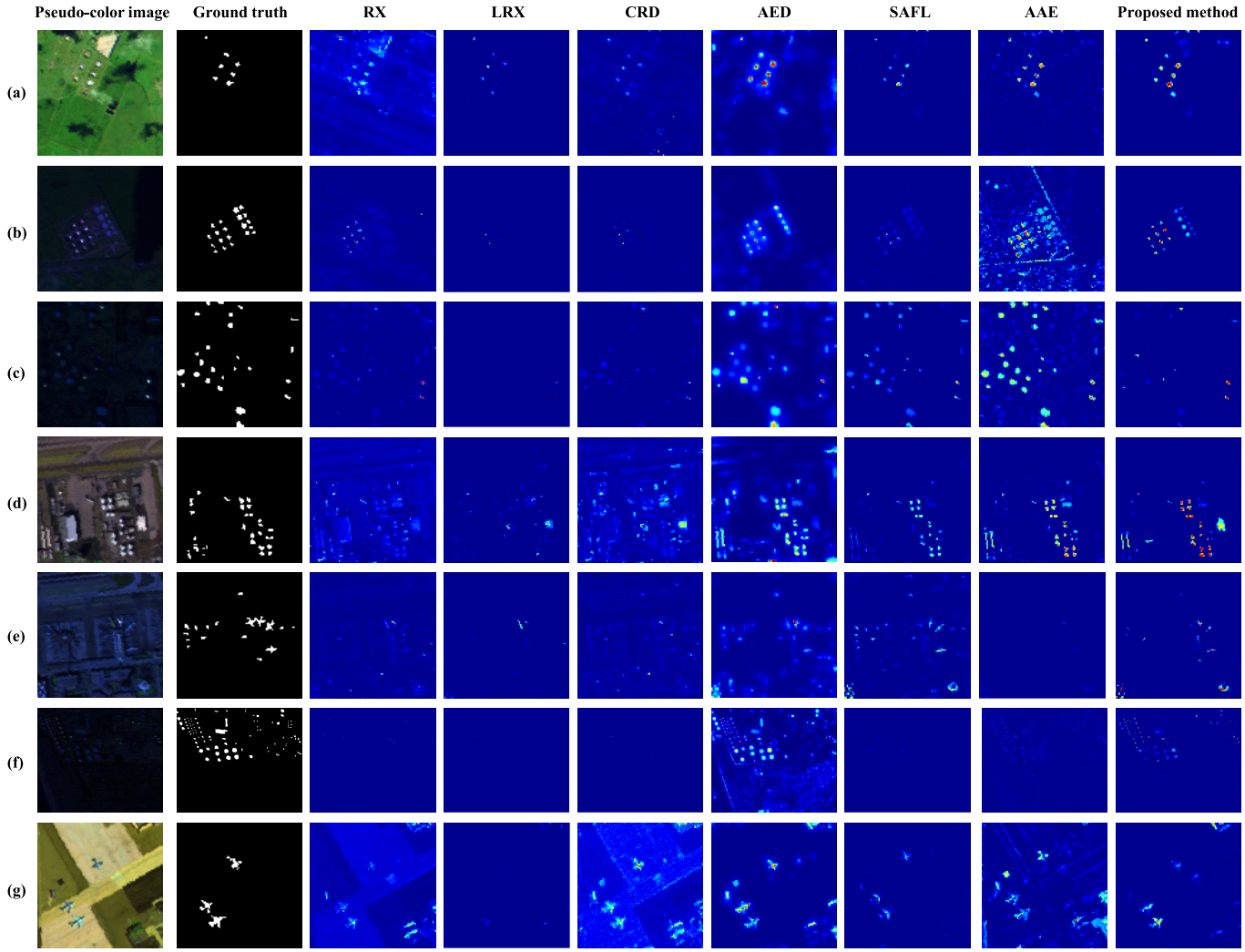


Fig. 5. Pseudocolor image, ground truth, and detection maps of the compared methods for (a) Texas Coast-1, (b) Texas Coast-2, (c) Los Angeles-1, (d) Los Angeles-2, (e) Los Angeles-3, (f) El Segundo, and (g) San Diego.

be detected. The pseudocolor image and the reference map are shown in Fig. 5(f).

4) San Diego Data Set: This data set was captured by the AVIRIS sensor covering the area of San Diego, which has 100×100 pixels and 189 spectral bands varying from 370 to 2510 nm. Three airplanes in this scene are regarded as anomalies. Hangars, parking aprons, and exposed soil make up the background. A series of background objects (hangar) have similar materials with the anomalies (airplane), which may result in difficulties in identifying anomalies from their surroundings. The pseudocolor image and reference map are shown in Fig. 5(g).

B. Experimental Setup

1) Comparison Methods: In this section, six popular anomaly detection methods are adopted as a comparison: RX, LRX, CRD, AED, SAFL [21], and AAE [21]. The RX detector, which calculates the squared Mahalanobis distance between a test pixel and the image background, is a widely used detector in the unsupervised anomaly detection field. The LRX, which detects anomalies by calculating the distance between a test pixel and the local image background, is an improvement version of RX. The CRD uses neighboring pixels for collaborative representation to achieve anomaly detection. The AED method is a recently developed method that has an outstanding detection performance on several real hyperspectral data sets. SAFL and AAE are two new deep learning-based anomaly detection algorithms in [21].

The inner window size in the LRX and CRD method w_{in} is varying from 3 to 19, and the outer window size w_{out} is varying from 5 to 23. The regularization parameter λ in the CRD method is fixed as 10^{-6}. For the AED method, the parameter setting is consistent with the article [18]. The parameter settings of SAFL and AAE are consistent with [21].

2) Performance Evaluation: In this experiment, the receiver operating characteristic (ROC) [50] curve and the area under the ROC curve (AUC) [51] is adopted to qualitatively and quantitatively evaluate the performance of the proposed method.

The ROC is a widely used evaluation metric in anomaly detection, which describes the varying relationship between

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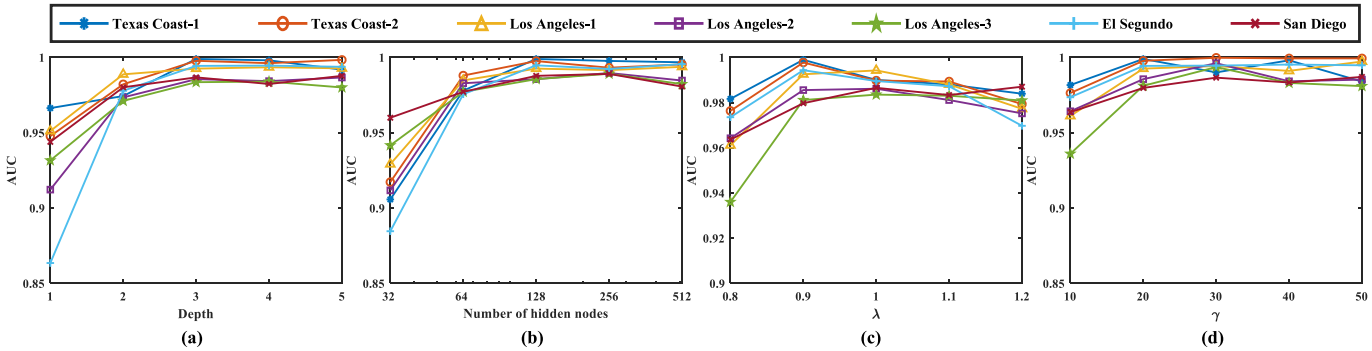


Fig. 6. Parameter sensitivity analysis of the influence on different data sets. (a) Depth. (b) Number of hidden nodes. (c) λ. (d) γ.

the true-positive rate (TPR) and false-positive rate (FPR) based on the reference map, and can be estimated via plotting the TPR and FPR at various thresholds τ. The AUC metric, which is derived from the ROC curve by calculating the whole area under the ROC curve, is regarded as a complete description of detection performance.

C. Parameter Sensitivity

This section aims to analyze the influence of different parameters on the performance of the proposed method. The proposed method has four main parameters, i.e., the depth D and the number of hidden nodes N of each subneural network, and the weighted parameters, λ and γ. The AUC scores of (TPR and FPR) are adopted to evaluate the object performance of the proposed method under different parameter settings. When one of the parameters is being analyzed, the other parameters are set as default values of $D = 3$, $N = 128$, $\lambda = 1.0$, and $\gamma = 30$. As shown in Fig. 6, with the increase in the value of parameter D, the AUCs tend to increase, while the AUCs become stable when the depth D is greater than 3. Similarly, it can be seen that when the number of hidden nodes N is greater than 128, the AUCs become stable and even decrease for some data set. In addition, the larger the scale of the network, the longer the computation time. Thus, the optimal N and D should be 128 and 3, respectively. As discussed in Section III, the weighted parameter γ balances the influence of the MSE and the discriminator features residual during the training procedure. As shown in Fig. 6, when the weighted parameter γ is greater than 30, the AUCs become stable. When λ is quite large, the detection accuracy is quite low since the network is underfitting. Meanwhile, when λ is quite small, the network overfits the training data set and has a weak generalization performance. Fig. 6 shows that when λ is equal to 0.9, the model has an optimal performance. Thus, for the testing data set, the optimal D, N, λ, and γ are 3, 128, 0.9, and 30, respectively.

D. Detection Performance

For the four real hyperspectral data sets, the detection maps, ROC curves, AUC scores, and the computing time of the compared anomaly detection methods are shown in Figs. 5 and 7 and Tables II and IV, respectively. It can be observed that the proposed method shows satisfactory performances and can achieve the highest AUC of (TPR and FPR) on average.

TABLE I
INFORMATION OF THE DATA SET

Name	Capture place	Resolution
Texas Coast-1	Texas Coast	17.2m
Texas Coast-2	Texas Coast	17.2m
Los Angeles-1	Los Angeles	7.1m
Los Angeles-2	Los Angeles	7.1m
Los Angeles-3	Los Angeles	7.1m
El Segundo	El Segundo	7.1m
San Diego	San Diego	3.5m

For the Texas Coast data set shown in Fig. 5(a) and (b), the RX method can fully detect the anomalies, while a series of strips noises, which corrupt the HSIs, are detected as false alarms. The AED method can provide a distinguished detection result and preserve the shape of the targets with a high detection rate, while its detection results exhibit a high false alarm rate for the Texas Coast-1 shown in Fig. 5(a) and miss some small targets for the Texas Coast-2 shown in Fig. 5(b). Similarly, the AAE method provides a more distinguished detection result but suffers from false alarms, as shown in Fig. 5(b). Although the LRX and CRD yield fewer false alarms, they miss a few anomalies and provide a low rate of detection. The SAFL method and the proposed method, which detect all the anomalies with a high detection rate and a low false alarm rate, acquire satisfactory detection performances compared with others. Meanwhile, the proposed method provides the most distinguished detection result that is closest to the ground truth. As shown in Tables II and III, the proposed method can achieve the highest AUC of (TPR and FPR) on average. The ROC curves are shown in Fig. 7(a) and (b).

For the Los Angles data set, detection results are illustrated in Fig. 5(c)–(e), and the LRX and CRD fail to preserve the shape of targets and miss some anomalies. For the Los Angles-1, the AED and AAE provide more distinguished detection results but suffer from false alarms. The RX, SAFL, and the proposed method can fully detect the anomalies with few false alarms compared with others, while the SAFL method provides the most distinguished detection result.

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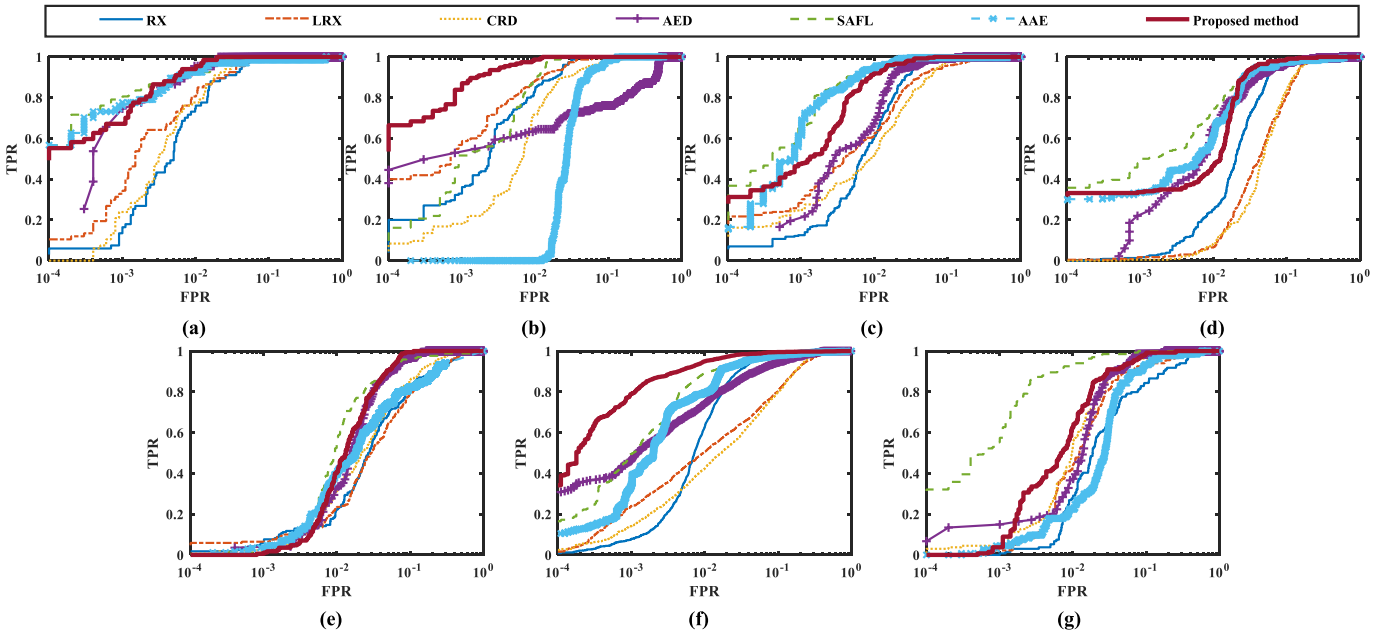


Fig. 7. ROC curves of the algorithms for (a) Texas Coast-1, (b) Texas Coast-2, (c) Los Angeles-1, (d) Los Angeles-2, (e) Los Angeles-3, (f) El Segundo, and (g) San Diego.

TABLE II
AUC SCORES OF (TPR AND FPR) OF THE ALGORITHMS FOR THE EXPERIMENTAL DATA SETS

Name	RX	LRX	CRD	AED	SAFL	AAE	Proposed method
Texas Coast-1	0.9907	0.9926	0.9929	0.9979	0.9973	0.9892	**0.9984**
Texas Coast-2	0.9946	0.9954	0.9862	0.9134	0.9962	0.9681	**0.9993**
Los Angeles-1	0.9887	0.9800	0.9806	0.9913	**0.9968**	0.9949	0.9963
Los Angeles-2	0.9692	0.9395	0.9404	0.9845	**0.9891**	0.9802	0.9862
Los Angeles-3	0.9288	0.9467	0.9564	0.9760	0.9713	0.9220	**0.9798**
El Segundo	0.9841	0.9444	0.9440	0.9804	0.9859	0.9875	**0.9962**
San Diego	0.9403	0.9678	0.9781	0.9815	**0.9971**	0.9595	0.9849
Average	0.9709	0.9666	0.9684	0.9750	0.9905	0.9716	**0.9916**

TABLE III
AUC SCORES OF (FPR AND τ) OF THE ALGORITHMS FOR THE EXPERIMENTAL DATA SETS

Name	RX	LRX	CRD	AED	SAFL	AAE	Proposed method
Texas Coast-1	0.0565	0.0049	0.0184	0.0239	**0.0030**	0.0094	0.0056
Texas Coast-2	0.0145	**0.0015**	0.0071	0.0160	0.0033	0.0423	0.0016
Los Angeles-1	0.0124	**0.0015**	0.0056	0.0314	0.0032	0.0203	0.0017
Los Angeles-2	0.0447	0.0076	0.0479	0.0316	**0.0034**	0.0092	0.0067
Los Angeles-3	0.0154	**0.0030**	0.0200	0.0224	0.0100	0.0136	0.0047
El Segundo	0.0027	0.0015	0.0031	0.0381	**0.0013**	0.0084	0.0015
San Diego	0.0599	0.0025	0.1087	0.0207	**0.0021**	0.0284	0.0063
Average	0.0294	**0.0032**	0.0301	0.0263	0.0038	0.0188	0.0040

For the Los Angles-2, the RX misses some targets. The AED can preserve the shape of the targets and provide a distinguished detection result but exhibit a high false alarm. The SAFL, AAE, and the proposed method can fully detect the anomalies with few false alarms, while the proposed method provides the most distinguished detection result that is closest to the ground truth. For the Los Angles-3, the AAE method fails to preserve the shape of targets and misses some anomalies. The SAFL and RX can detect all the anomalies but suffer from false alarms. The AED and the proposed method can fully detect the anomalies with few false alarms compared with others. As shown in Tables II and III, the proposed method achieves the second-highest AUC scores of (TPR and FPR) for Los Angles-1 and Los Angles-2, the highest AUC score of (TPR and FPR) for Los Angles-3, and the second-lowest AUC score of (FPR and τ). The ROC curves of the compared methods are shown in Fig. 7(c).

For the El Segundo data set, the detection results are shown in Fig. 5(f), and the RX and LRX methods have the ability to detect a few anomalies with low confidence of detection, but the shapes of some targets cannot be preserved. The CRD, SAFL, and AAE can identify the locations of the anomalies and better preserve the shape of the targets, while the separation between the anomalies and background is low. The AED almost detects all the anomalies and preserves the shape of the targets, but a few false anomalies are also detected. While the proposed method can generate a detection result with better background suppression. As shown in Table II, the proposed method achieves the highest AUC score of (TPR and FPR). The AUC scores of (TPR and FPR) of the RX, LRX, CRD, SAFL, AAE, and AED methods are 0.9841, 0.9444, 0.9440, 0.9859, 0.9875, and 0.9804, respectively. The ROC curves are shown in Fig. 7(f). Furthermore, the proposed method achieves a relatively low AUC score of (FPR and τ) of 0.0015.

For the San Diego data set, as shown in Fig. 5(g), the RX, CRD, AAE, and AED methods almost fully detected the anomalies but cannot suppress the background interference. The LRX method avoids false alarm but provides a low detection rate. The SAFL achieves the highest AUC score of (TPR and FPR) but cannot preserve the shape of the

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TABLE IV
COMPUTING TIME (SECONDS) OF DIFFERENT METHODS

Name	RX	LRX	CRD	AED	SAFL	AAE	Proposed method
Texas Coast-1	**0.22**	120.03	414.90	**0.22**	105.98	104.83	9.77
Texas Coast-2	**0.21**	123.34	374.69	0.28	20.33	19.54	8.96
Los Angeles-1	**0.24**	123.83	405.95	0.29	44.45	44.21	9.43
Los Angeles-2	**0.29**	125.73	382.91	**0.29**	25.91	26.03	10.95
Los Angeles-3	**0.23**	122.33	396.15	0.28	55.30	67.77	8.98
El Segundo	**0.40**	497.39	2776.22	7.52	271.03	269.62	56.88
San Diego	**0.25**	120.12	412.31	0.27	43.98	41.71	9.75

targets. The proposed method preserves the shape of the targets properly and achieves a second-highest detection AUC score of (TPR and FPR). The AUC score of (TPR and FPR), the AUC score of (FPR and τ), and the ROC curves are provided in Tables II and III and Fig. 7(g), respectively.

In order to evaluate the computational complexity, as shown in Table IV, we further tabulate the actual runtime of different methods. Although the proposed method is not the most efficient method, it is much faster than the LRX, CRD, SAFL, AAE, and AED methods. All the experiments have been carried out on an Intel Core-i5 PC with 8 GB of RAM.

V. CONCLUSION

In this article, a novel SSL framework is proposed for HSI anomaly detection by exploring both spectral and spatial features. First, a GAN-based model is proposed to estimate the background distribution in a semisupervised manner and an initial spectral feature in which each pixel represents that an anomaly likelihood can be inferred by applying the trained model to all the spectral vectors. Then, a morphological attribute filter is adopted to generate an initial feature in the spatial domain. Finally, a feature fusion technique is used to suppress the background and combine the complementary information in different aspects to obtain a fused detection map. A comparison has been done with well-known HSI anomaly detection methods. Experiments performed on various real HSIs demonstrate that the proposed method can outperform in terms of quantitative evaluation, visual comparison, and computationally effectiveness. Furthermore, generalizing the proposed method to real applications and considering some more reasonable criteria to evaluate anomaly detection will be the focus of our future work.

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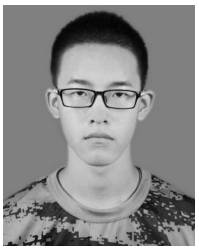
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